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Nash Equilibrium

Using Beamer

Let's Start with a Story

HONDA

TOYOTA

Table

Profit IN Millions		TOYOTA Company	
		Building new plant	Not Building plant
HONDA	Building new plant	16 16	
	Not Building new plant		

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Profit IN Millions		TOYOTA Company	
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Guess,Who will be the winner?

Before ending of this presentation,we will be able to know the answer

Introduction to Nash Equilibrium

The concept of Nash Equilibrium was created by John Nash, an American mathematician.

Introduction to Nash Equilibrium

John Nash was awarded the Nobel Memorial Prize in Economic Sciences in 1994 for his contributions to game theory, including the concept of Nash Equilibrium.

Introduction to Nash Equilibrium

Nash Equilibrium is a concept in game theory where each player chooses the best strategy, considering the strategies of others. It occurs when no player can improve their outcome by changing their own strategy, as long as the others stick to theirs. This idea helps to understand decision-making in situations where people or companies compete or cooperate. Nash Equilibrium explains how participants often reach a stable outcome even without directly communicating.

Why Nash Equilibrium ?

Why Nash Equilibrium ?

Now I can take best decisions

Why Nash Equilibrium ?

- people make decisions
- predicting outcomes
- stable and predictable results.
- everyone could do better if they cooperated
- decision-making with interdependence
- navigate conflicts and cooperation.

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There are eight types of Nash Equilibrium based on different classifications in game theory.

We will discuss all categories of Nash Equilibrium. However, the first two types are the core categories, so we will talk about them in detail later.

Types of Nash Equilibrium

- Pure Strategy Nash Equilibrium
- Mixed Strategy Nash Equilibrium
- Symmetric Nash Equilibrium
- Asymmetric Nash Equilibrium
- Strict Nash Equilibrium
- Weak Nash Equilibrium
- Subgame Perfect Nash Equilibrium (SPNE)
- Correlated Nash Equilibrium

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Pure Strategy Nash Equilibrium

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- No player can improve their outcome by unilaterally changing their strategy.
- Maximizes their payoff given the strategies of others.

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		Loyal	Fink
Player 1	Loyal	-1 -1	-3 0
	Fink	0 -3	-2 -2

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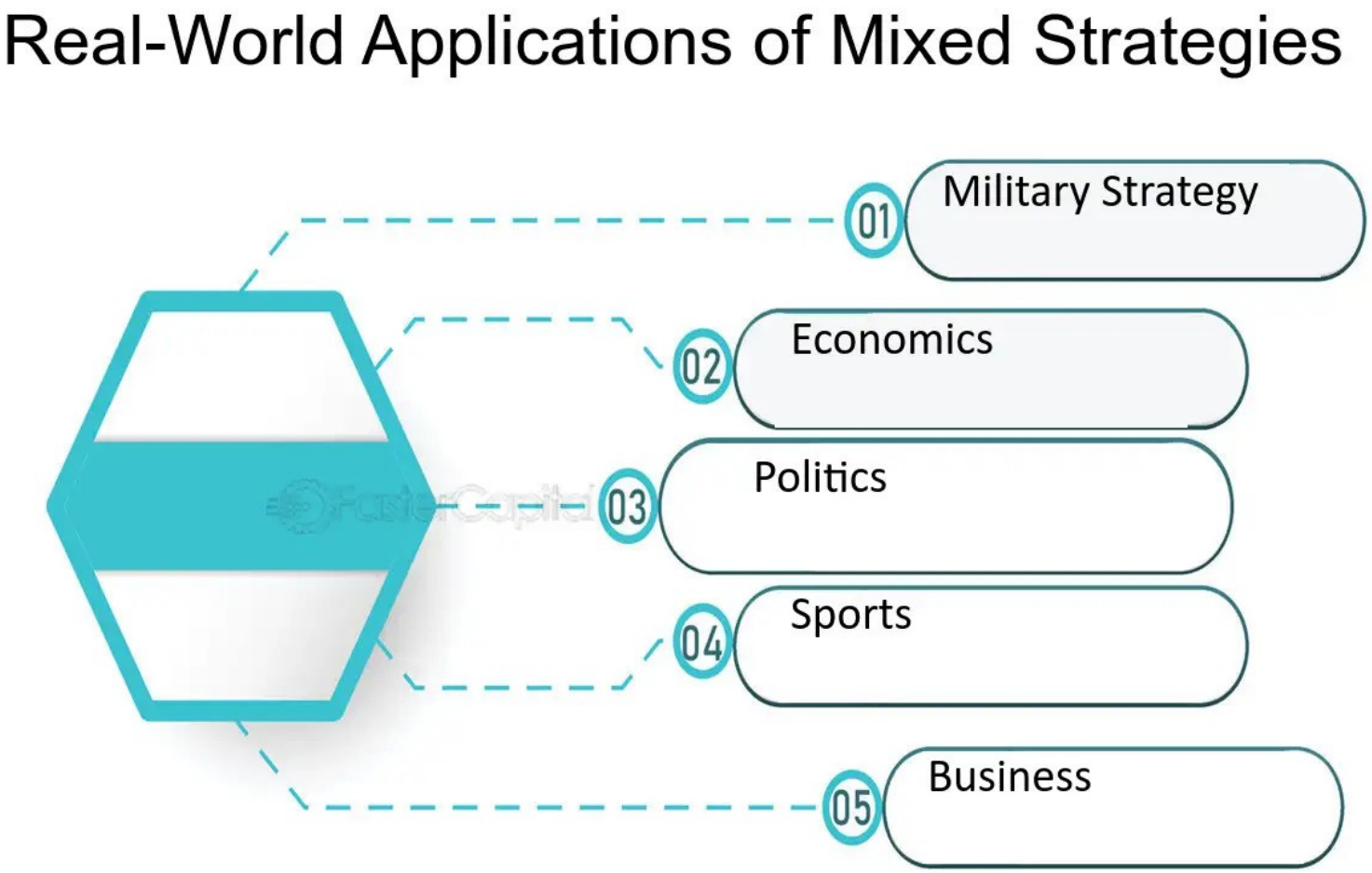
Mixed Nash Equilibrium Example